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AI Regulates AI

An Artifact for Automated Risk Assessment

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Abstract: The use of AI systems in organizations is growing rapidly and must be managed responsibly. The EU Artificial Intelligence Act (AIA), the world's first comprehensive AI law, requires organizations to ensure responsible AI deployment by classifying their systems based on risk levels. This paper adopts a Design Science Research (DSR) approach to develop an artifact for the automated risk classification of AI systems. We performed a systematic literature review and conducted expert interviews to identify current challenges in risk level classification according to the AIA. Based on our findings, we developed a multi-agent AI system that performs automatic risk classification and generates natural language output. This backend system can be accessed via a web user interface. Our prototype achieves a classification accuracy of 88% across all risk levels and offers small and medium-sized enterprises an accessible way to integrate AI compliance into their processes.

Keywords: Generative Artificial Intelligence, EU Artificial Intelligence Act, AI Agents, Design Science Research

1 Introduction

The use of AI systems was previously reserved for large companies, but this is now changing due to the easy availability to Generative Artificial Intelligence (GenAI), which is now easily available to small and medium-sized enterprises (SMEs) [Ri22]. This opens up new opportunities for SMEs to assert themselves in today's competitive environment and to gain strategic advantages [NP24]. For instance, GenAI use can optimize internal business processes e.g., decision-support systems for hiring [Ch24] (positive effect). However, negative side-effects such as increased algorithmic bias, which may lead to unfair candidate selection, can occur simultaneously [Mb24]. This tension has potentially

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profound societal implications [Fl18] and highlights the increasing urgency of effective AI governance aimed at accountability, fairness and transparency [Ca18]. Therefore, the European Commission (EU) adopted the EU Artificial Intelligence Act (AIA) in 2024 [Th24]. AIA is the world's first comprehensive AI regulation and sends an important signal for the responsible use of AI [Eu23]. Based on a risk-based approach, companies must categorize their AI systems into four risk levels: unacceptable risk, high risk, limited risk, and minimal risk. Consequently, the first necessary step towards compliance with AIA and its implementation is the correct classification of risk level. For many SMEs that lack legal expertise, these requirements can cause a significant challenge [So25]. Against this backdrop, we have developed an automated approach that uses AI to enable systematic risk classification in accordance with the AIA. Moreover, an AI Catalog is created, in which all AI systems can be documented and readily accessible according to their risk level. This approach can facilitate compliance activities within the organization. Therefore, we seek to contribute to both theory and practice by answering the following research question (RQ): *How can SMEs handle the challenges of correct risk classification under the AIA?* To answer the RQ, we apply an Design Science Research (DSR) approach adapted from Hevner [He07] to design and develop an information system, which is implemented and evaluated as a proof of concept. Based on a systematic literature review adapted from Vom Brocke et al. [Vo09, Vo15b], we identified the underlying problem and built a theoretical knowledge base. Using the Delphi method adapted from Skulmoski [SHK07] we integrated expert knowledge. Their expertise allowed us to identify challenges in the implementation of the AIA and to derive concrete requirements for our artifact. On this basis, we were able to create an artifact that combines theoretical and practical requirements. We make the following main contributions. First, we present an information system named Compliance Tool that supports AI risk classification under the AIA using GenAI. Second, we implement it as a multi-agent system combining a ReAct-based interface, retrieval-augmented generation (RAG), and semantic search (FAISS) for regulation-aligned, context-aware classification with natural language output. Third, the results are stored in an AI catalog and can be reviewed and adjusted via a human-in-the-loop mechanism to ensure quality.

2 Conceptual Foundations

2.1 Large Language Models and Retrieval-Augmented Generation

Large Language Models (LLMs) are known for their impressive performance in handling a wide range of natural language tasks through prompts [Le23]. One of the most well-known LLMs is ChatGPT, which has been trained on extremely large datasets and demonstrates remarkable capabilities in language understanding and logical reasoning [Br20]. However, a key limitation is the outdated nature of the underlying knowledge, especially in rapidly evolving domains like legal frameworks. This is especially relevant for our study, which aims to enable SMEs to automate the classification according to AIA using an AI system. In this context, pretrained models face inherent limitations. Therefore, the use of RAG is recommended. RAG combines an external knowledge source with an LLM to dynamically incorporate current, domain-specific knowledge into the response generation process [Wi24]. There are various ways to implement RAG to ensure that relevant content from external sources can be efficiently retrieved. For example, documents can be divided into smaller semantic units such as paragraphs and transformed into vectors using embedding models [Mi13]. These vectors represent the semantic content of a text passage in multidimensional space and are stored in a vector database. The open-source library FAISS (Facebook AI Similarity Search), developed by Meta, can be used for this purpose. It enables efficient similarity searches and clustering of dense vectors [BJ25]. LangChain has established itself as a helpful open-source framework for coordinating the components in RAG-based architectures [La25]. It was designed to simplify the integration of LLMs into applications by connecting a modular chain of components via a unified API. In addition, LangChain allows the flexible integration of agents that handle core tasks such as routing queries, selecting appropriate knowledge sources and coordinating individual processing steps [YZ24].

2.2 Overview of the AIA and its application in SMEs

Adopted in 2024, the AIA is the world's first comprehensive legal framework for AI [EU23]. It aims to ensure safe, transparent, and fair AI deployment while maintaining human oversight. Based on a risk-based classification, the AIA defines four risk levels: unacceptable, high, limited and minimal. Unacceptable-risk systems, such as social scoring or real-time biometric surveillance in public spaces, are strictly prohibited [Th24, Chap. II]. High-risk systems, which may impact health, safety, or fundamental rights, must meet strict requirements, including risk management, high-quality training data, documentation, logging, user transparency, human oversight and robustness [Th24, Chap. III]. Limited-risk systems are subject to specific transparency obligations, such as labeling AI-generated content [Th24, Chap. IV]. Minimal-risk systems, like spam filters, pose negligible harm and are exempt from regulation [Th25]. The AIA places particular emphasis on supporting SMEs, for which full compliance with regulatory requirements can pose a significant challenge. To address this, the regulation provides simplified documentation requirements, reduced fees and prioritized access to so-called regulatory sandboxes. These sandboxes enable SMEs to develop and test innovative AI solutions under real-world conditions, regardless of the associated risk level. Despite these supportive measures, systematic risk classification remains essential to ensure regulatory compliance with the AIA [Li22]. Previous research has addressed different challenges associated with risk classification. Hanif et al. [Ha23b] propose manually structured decision trees, while Mustroph and Rinderle-Ma [MR24] introduce a prototype software-as-a-service (SaaS) quality management system that supports classification, data governance and risk management. In contrast, our artifact follows a fully automated approach. A theory- and expert-driven multi-agent system leverages the AIA to dynamically and compliantly classify risks and persistently store the results in a database.

3 Method

We follow a DSR approach (Figure 1) according to Hevner [He07]. The outcome of this method is an artifact, in our case an Compliance Tool for the classification of AI systems within the context of the AIA. An artifact represents a generalizable design principle rather than a final solution [He07, Vo15a]. Hevner's framework consists of three interconnected cycles: the Relevance Cycle, which links the application domain with design activities, the Rigor Cycle, which integrates theoretical grounding and scientific rigor into the design process and the Design Cycle, which iteratively builds and evaluates the artifact, together form the three core components of the DSR framework.

Aforementioned, to fullfill the **Rigor Cycle**, we reviewed the literature in accordance with the principles of a systematic literature review [WW02]. Accordingly, we conducted a systematic literature research, adapting the steps proposed by Vom Brocke et al. [Vo09, Vo15b] in three steps. We defined a concept matric [WW02] in Step 1, to categorize the problems resulted in adapting the AIA in organizations. Step 2 focused on populating the matrix with relevant literature findings, guided by four inclusion criteria to ensure the relevance and quality of the selected publications. First, we included only empirical literature that addresses the classification of AI systems in organizational contexts in relation to the AIA. Second, we included only literature published since 2021, as the European Commission presented its proposal for an Artificial Intelligence Act on April 21, 2021 [Th24]. Third, to enhance academic rigor, only peer-reviewed publications were considered. Fourth, for reasons of clarity and accessibility, we considered only English-language publications. After defining these inclusion criteria, we based our database selection on Gusenbauer and Haddaway's [GH19] evaluation study, identifying SCOPUS as the most suitable platform due to its extensive coverage, reproducibility and flexible search functionalities. The second activity within this step involved designing and conducting a SCOPUS query. The query string applied titles, abstracts and keywords was as follows: *TITLE-ABS-KEY (("generative artificial intelligence" OR "generative ai" OR "genai" OR "gai" OR " Artificial Intelligence" OR "AI") AND ("The EU Artificial Intelligence Act" OR "European Union AI Act" OR "EU AI Act")) AND PUBYEAR > 2021*. We executed the query on November 31, 2024, resulting in 224 publications. After screening titles and abstracts, the selection was narrowed down to 44 works. A full-text analysis of these 44 publications identified six studies that met all of the predefined inclusion criteria. This activity was conducted by two authors, and any differences in classification were resolved through discussion until consensus was reached. In Step 3, we applied axial coding [CS90] to the extracted findings within each category of the concept matrix in order to abstract the theoretical grounding, scientific foundations and problem relevance.

The activites within the **Relevance Cycle** are not only to identify the underlying problem but also to find the acceptance for our Compliance Tool. To gather practical insights into the regulatory context, we applied the Delphi Method according to Skulmoski [SHK07]. Our research consists of two Delphi rounds and includes seven steps. First, we formulated the research question, which was derived from the outcomes of the systematic literature review in the Rigor Cycle. Accordingly, the goal of the expert interviews was to identify the challenges that arise in complying with the AIA, particularly regarding risk classification, as well as to understand the technical requirements the tool must meet to support AI compliance in organizations. Second, we designed the research instrument. To answer the research question, we decided on a qualitative research design in the form of expert interviews and an online survey. Third, we used a research sample of nine experts from business units in the automotive and mechanical engineering supply industry, IT management consulting and conglomerates. Their organizations are affected by the AIA and therefore capable of answering our research question. Fourth, we developed the questionnaire and discussed it based on the findings in the Rigor Cycle. Fifth, we conducted the expert interviews in individual online meetings to ensure participant anonymity. Sixth, we developed the second Delphi round, prepared the results of the first round, and presented them to the experts. In this manner, they were given the opportunity to reflect on, revise, or expand their responses. Following this, we selected an online survey to specify the artifact requirements. Seventh, we verified, generalized and documented the research results, which served as the basis for the design of our artifact.

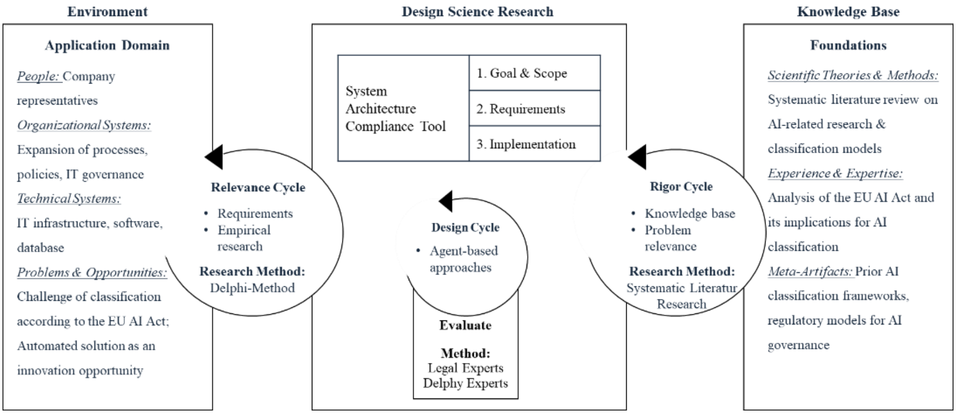

Fig. 1: Based on DSR by Hevner with the design adapted from Hagenhoff et al. [Ha24]

In the third step of DSR, we applied the results from the Rigor and Relevance Cycles to the development of our system architecture for our Compliance Tool within the **Design Cycle**. First, we defined the objective and scope of our system architecture as the starting point for the development process. Second, we mapped the requirements identified in the second Delphi round to our architecture. Third, we implemented a modular system architecture that integrates semantic retrieval and automated risk classification to support compliance with the AIA. Fourth, we evaluated the artifact. First, the experts participate in a third Delphi round, where they tested and evaluated the compliance tool via an online survey. Second, we tested the classification accuracy using 20 fictional use cases, which were validated by a legal expert. The following section summarizes the core aspects of our approach derived from all three cycles.

4 Results

4.1 Problems and requirements definition

The literature review identifies six relevant sources: A precise description of the AI system is fundamental for correct risk classification [To22], yet this is often hindered by a lack of resources or expertise. Documenting the data used, such as for tracing potential bias or discrimination, also challenging, as human judgment alone can be prone to error [Br22]. Moreover, companies struggle to clearly assign their systems to a specific risk category under the AIA. This is partly due to the ambiguous definition of the high-risk category, which may lead to over- or underregulation [Ha23a]. However, precise classification remains a prerequisite for regulatory compliance [Mi24]. In addition, there is a lack of methodological guidelines [Eb24] and practical tools to support implementation [MR24]. Based on the literature, interview questions were developed to address internal corporate challenges and requirements for our Compliance Tool for classification in accordance with the AIA. Nine experts participated in the interviews. The experts reported difficulties in interpreting the legal framework and integrating regulatory requirements into existing processes. In particular, documentation and assessment were described as time-consuming and prone to error. The experts emphasized the need for automation and consistent classification. There is a lack of scalable, standardized solutions that allow for dynamic assessments and can be integrated into existing workflows. Compliance must be understood as an integral part of AI development. In addition, the participants expressed a need for centralized platforms for monitoring, documentation and real-time updates. Transparency and explainability were described as crucial for building trust, which is difficult to implement. In the second round of the Delphi study, the requirements for the artifact were consolidated based on the results of the first round. The experts were asked to evaluate five requirements on a scale ranging from ‘not at all’ [0], ‘slightly’ [1], ‘moderately’ [2], ‘considerably’ [3], to ‘completely’ [4]. The ability to store information about an AI system in a database and reuse it for classification purposes was rated as highly important (requirement 1; mean = 3.88). Closely related is the artifact’s ability to suggest a risk level in accordance with the four categories defined by the AIA (requirement 2; mean = 3.50). In addition, the integration of a human-in-the-loop mechanism was emphasized to validate the suggested risk level before final classification (requirement 3; mean = 3.38). Following this, the final classification should be stored automatically in the system to efficiently document the compliance process (requirement 4; mean = 3.13). Traceability was also highlighted: the artifact should be able to explain the classification decision in an understandable manner (requirement 5; mean = 3.38). Our findings show a strong alignment between the results of the systematic literature review and the expert interviews, indicating that similar challenges were identified in both theoretical and practical contexts.

4.2 System Architecture

The system architecture of our Compliance Tool (see Figure 2) is divided into three core components: Frontend, AI Middleware and Backend. Usage occurs in two separate steps, allowing tasks to be distributed flexibly between different users. For example, one user maintains the AI Catalog, while another performs the risk classification. In the first step (1), the AI Catalog interface serves as the central frontend element, where users can edit, delete, or create entries for AI systems (requirement 1). For this purpose, seven fields must be filled. These fields are generated through an AI based structured multi-agent approach and cover system identity (name, description), application domain (sector, goal), technical specifications (input and output data) and an indication of whether sensitive data is processed according to Article 9 of the General Data Protection Regulation (yes/no). This structured metadata enables systematic documentation, analysis and subsequent classification of AI systems. The entries are then stored in a SQLite database (2) by the AI Catalog Writer Agent, which serves as a central repository for regulatory assessments, risk classifications and compliance purposes. In the second step, the user interacts with the system via a natural language chatbot interface (3) for instance by asking a question such as "Make a suggestion for the risk level of the AI system called *name*" (requirement 2).

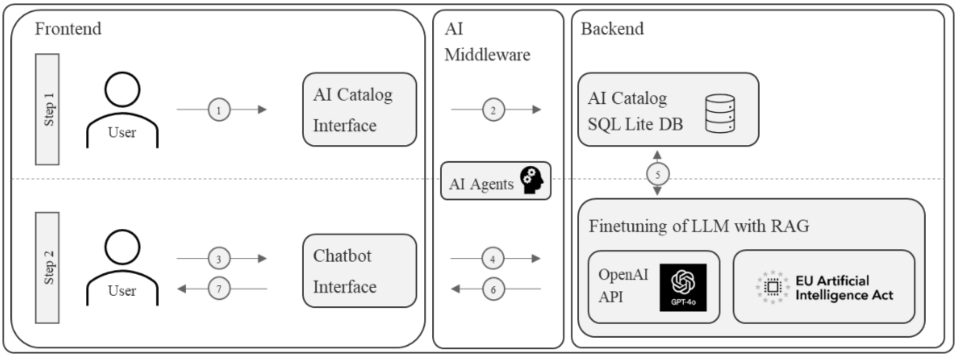

Fig. 2: System Architecture of the Compliance Tool

The main ReAct Agent interprets the request (4) and forwards it to the AI Agents (Risk Classification Agent and AI Catalog Writer Agent). These three agents form a modular multi-agent architecture, based on the LangChain framework. The underlying large language model is GPT-4o, integrated via the OpenAI API. A RAG approach is used to integrate regulatory knowledge. This involves transforming the AIA from PDF format into semantic text segments, which are then converted into high-dimensional vectors using OpenAI's pretrained embedding model. These vectors are stored in FAISS. Then, the Risk Classification Agent accesses the metadata stored in the AI Catalog (5) and applies a decision tree developed with ChatGPT, based on the classification criteria defined in the AIA. This enables the properties of the requested AI system to be aligned with the regulatory classification criteria and the system to be categorized accordingly. In parallel, the AI Catalog Writer Agent updates the entry in the AI Catalog (5) with the determined risk level (requirement 4). Then, the main ReAct Agent collects the classification result (6), which is returned to the user via the frontend (7). The user may optionally involve a human-in-the-loop, who can revise the classification if the automated result is incorrect (requirement 3). The main ReAct Agent explains how the result was reached (requirement 5). For example, this explanation may state: “The suggested risk level for the AI system *name* is ‘unacceptable’. This is based on the AIA’s risk classification, which considers factors such as the involvement in biometric identification or categorization of individuals”.

4.3 Evaluation

First. As a formative step within DSR, we conducted an early expert evaluation of the implemented requirements. The experts received an offline demo of our compliance tool. This demo showcased all steps as described in Chapter 4.2. Using an online survey, the experts were asked to assess the model on a five-point scale ranging from ‘not at all’ [0], ‘slightly’ [1], ‘moderately’ [2], ‘considerably’ [3] and ‘completely’ [4]. As the evaluation was based on voluntary participation, only four of the initial nine experts took part. The highest agreement scores were given for the automatic storage of the risk classification (mean = 3.75), the explanation of the classification decision (mean = 3.75) and the suggestion of a risk category in accordance with the AIA (mean = 3.5). To answer legal questions and the data entry process in the AI Catalog were also rated positively (mean = 3.25). Lower scores were given for the human-in-the-loop feature (mean = 2.75). In the qualitative feedback, the experts emphasized the need for more transparency and legal traceability, such as referencing legal texts and marking unclear cases for human review.
Second. Our model was tested using 20 fictional AI systems generated by ChatGPT, with five systems assigned to each of the four AIA risk levels. For this purpose, we conducted testing cycles over three consecutive days (in sum 60 samples). Tabel 1 compares the actual classification of the 60 samples with the classification by the Compliance Tool using a a two-dimensional matrix with four classes: minimal, limited, high und unacceptable. The main diagonal shows the respective number of true positives (TP - correctly classified) per class in bold. The true positive rate (TPR) or recall is the number of correctly classified AI systems (TP) per class in relation to the number of actual AI systems per class. Precision is the ratio of the TP per class to the number of AI systems classified by the Compliance Tool per class [GKG15]. The accuracy shows the total number of correct classifications in relation to all classifications. The analysis of the classification results shows an overall high accuracy of 88%. The TPR is high across most categories, although systematic deviations were observed. Minimal risk was correctly classified in all 15 cases. Both the TPR and precision are 100%, indicating highly reliable detection. Limited risk was correctly identified in 87% of cases. The remaining 13% were classified as high risk, which suggests a tendency toward conservative classification. Nevertheless, the precision for this class was also 100%. High risk achieved a true positive rate of 100%, but its precision is only 68%. This means that 32% of the systems classified

as high risk are actually assigned to another category. Specifically, this reflects an overestimation of five unacceptable risk cases. Unacceptable risk category showed the lowest TRP at only 67%. In 33% of our samples for this category, systems that should have been classified as unacceptable were instead labeled as high risk, indicating that critical cases were underestimated by one risk level. However, the precision for this category was 100%. All systems classified as unacceptable are correctly classified.

		Actual classification level					
		minimal	limited	high	unacceptable	∑	precision
Classification Compliance Tool	minimal	**15**	0	0	0	15	1.00
	limited	0	**13**	0	0	13	1.00
	high	0	2	**15**	5	22	0.68
	unacceptable	0	0	0	**10**	10	1.00
	∑	15	15	15	15	60	
	TPR	1.00	0.87	1.00	0.67		
	accuracy						0.88

Tab. 1: Classification of AI Systems by the Compliance Tool

5 Discussion

Our results are threefold. First, using DSR, we developed an artifact in the form of an information system. It systematically combines theoretical foundations and challenges discussed in the scientific community regarding the classification under the AIA with practical requirements identified through expert interviews. Second, we designed the artifact with an interface that enables risk classification of AI systems using natural language. Therefore, we use a multi-agent framework. The integration of FAISS and a RAG approach enhances the semantic precision and alignment of the classification with regulatory requirements. This methodological approach could also be adapted to automate compliance processes for other regulatory frameworks beyond AIA. Third, the developed system provides concrete support for companies of all sizes in implementing risk classification requirements. It can be integrated early in the AI system development process to offer initial indications of regulatory obligations. Since our AI Compliance Tool does not achieve 100% classification accuracy and shows a tendency to misclassify high-risk systems, a human-in-the-loop mechanism is built in. This allows users to efficiently review and, if necessary, adjust system entries via the AI Catalog interface. Thus, the system facilitates a low-threshold first step toward achieving compliance with AIA.
Our work has three main limitations. First, the decision tree used in our system is based on a structure generated by ChatGPT. Existing validated decision trees (e.g., Hanif et al. [Ha23b]) were not integrated. Future research should systematically incorporate such validated decision structures and evaluate whether this improves classification accuracy. Second, our evaluation was conducted with only four instead of nine experts and based solely on synthetic test data. Since real-world AI systems can vary significantly in

structure and level of detail, validation with real data is necessary to assess the generalizability and practical applicability of the system. Third, we relied exclusively on FAISS as the vector database. Future research could explore the use of other RAG approaches (e.g., graph databases) to further enhance performance and flexibility. Moreover, Kott et al. [Ko25] highlight a paradoxical tension that remains unaddressed: While our classification tools can support regulatory compliance and increase the efficiency of organizational business processes, negative side effects can occur due to the energy consumption of inference. The potential benefits must always be weighed against the actual energy consumption in order to ensure that the issue of sustainability is not ignored. Therefore, we want to evaluate the energy consumption of similar AI systems and identify energy-efficient IT system architectures.

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